Comparing Costs and Utilization Between Provider Types for Back and Neck Pain: A Cross-Sectional Study

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Abstract

Objective: The purpose of this study was to compare treatment expenditures and utilization of licensed doctors of chiropractic or doctors of physical therapy working for Airrosti (AP) compared with non-AP providers (NAP) comprising 5 different provider types.

Methods: This study was a retrospective, claims-based, cross-sectional study using 5 years of claims and enrollment data from the state of Texas. We compared licensed doctors of chiropractic or doctors of physical therapy working for Airrosti to NAP. Episodes of care for back and neck pain were formed using 90-day clean periods. We utilized propensity scores using inverse probability weighting to control for selection bias. We examined the association of provider type with total costs per episode and 5 different measures of utilization: numbers of visits, length of episode, hospitalization, use of advanced diagnostic imaging, and use of surgery.

Results: Included were 645 799 unique patient episodes of care. Orthopedic specialists, physiatrists, and physical therapists had higher costs and utilization than AP, except for lower visits for orthopedic specialists. Primary care providers had lower costs and utilization than AP, except for hospitalizations, in which no difference was found. AP were most similar to chiropractors in terms of costs and utilization; however, AP had higher use of advanced diagnostic imaging compared with chiropractors. Chiropractors had more visits and longer episodes than AP. Standard deviations for average episode cost, episode length, and number of visits were greater for all provider types compared with AP, except for primary care providers.

Conclusion: In the sample studied, providers using standardized treatment pathways had reduced variation and costs for patients with spinal pain. We hypothesize that adherence to treatment pathways that align with recommended clinical practice guidelines that discourage the use of diagnostic imaging and surgery as a first step for treating lower back and neck pain may have resulted in the significant cost and utilization differences found between AP and NAP. (J Manipulative Physiol Ther 2025;00;1-10)

Key Indexing Terms: *Back pain; Neck pain; Episodes of care; Health care utilization; Health care costs; Chiropractic; Physical therapy specialty; Primary care; Orthopedics; Physiatry*

Introduction

Back and neck pain are leading causes of disability throughout the world.[1] In the United States, back and neck problems were reported to be the most common cause of disability among working-age adults, affecting 6.1 million people from 2011 to 2013.[2] The costs of treating these conditions are substantial and rising. In the United States, total annualized treatment costs exceeded $134.5 billion in 2016—the highest among 154 conditions examined,[3] and exceeded costs for diabetes and ischemic heart disease, which had the highest health care spending in 2013.[4]

Recent practice guidelines for low back pain emphasize the early use of nonpharmacologic care with limited use of diagnostic imaging and surgery.[5,6] Yet, large gaps persist between recommended guidelines and treatment practices.[5] These gaps contribute to wasted resources and unnecessary patient harm, such as increased exposure to radiation, increased risk of opioid addiction, overdose, death, and increased risk of complications and mortality from complex spinal surgeries.[5]

Contributing to practice variations are the many different provider types and treatment pathways involved in the care of back and neck conditions, resulting in varying costs and patterns of utilization.[7-9] These providers include

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physical therapists, chiropractors, orthopedic specialists, primary care providers, and physiatrists working in a variety of practice settings. However, in recent years, new business models and treatment approaches for musculoskeletal conditions have emerged.

One example is Airrosti Rehab Centers, which is a health care provider group that delivers musculoskeletal care through a standardized business model and clinic processes across over 150 practice locations in Texas, Washington, Virginia, Ohio, and the District of Columbia.[10] Though elements are common across chiropractic and physical therapy practices,[11,12] this company employs clinical and business processes to standardize care and reduce variability in outcomes and cost across its providers and practices (Supplemental Data).

The purpose of this study was to compare expenditures and utilization of licensed doctors of chiropractic or doctors of physical therapy who work for Airrosti (AP) with non-AP providers (NAP) across 5 provider types: physical therapists, chiropractors, orthopedic surgeons, primary care providers, and physiatrists. We hypothesized that there would be cost and utilization differences between AP and NAP. Examining these differences could contribute to the literature on the potential for standardized treatment pathways to reduce variation and costs for spinal pain.

Methods

Study Design

Our study used a cross-sectional study design using deidentified patient claims and enrollment data from a national commercial claims-based dataset. The insurer was a national commercial carrier that pays claims for the general public as well as serves as a third-party administrator for self-insured plans.

Setting

Airrosti is a San Antonio-based physical therapy for profit company that specializes in musculoskeletal care. The majority of clinics are located in Texas; therefore, the study sample was restricted to this state. The study period ranged from January 1, 2017, through December 31, 2021; however, claims from January 1, 2016, through March 31, 2021, were examined to identify diagnoses used in the propensity score (by provider type analysis), construct patient episodes, and determine exclusionary criteria.

Participants (Patient Episodes)

Patient episodes were the unit of analysis. Episodes were formed by retrospectively accumulating the bills associated with 1 or more claims falling within 90 days of the index and subsequent visits. A 90-day clean period was also enforced at the beginning and end of the study period, so that episodes with bill dates before January 1, 2017, or after December 31, 2021, were removed. Only patients' first episodes within the study period were included in the analysis.[7]

Claims with a primary International Classification of Diseases, 10th Edition (ICD-10), diagnosis code of back or neck conditions treatable by AP were included: a total of 462 codes from G43.119 (migraine with aura, intractable, without status migrainosus) to S39.013S (strain of muscle, fascia and tendon of pelvis, sequela). Though classified as a neurologic condition in the ICD, there is growing evidence of the effective use of nonpharmacologic options such as manual therapy in the management of migraines.[13-15]

Because Airrosti does not participate in standard worker's compensation relationships with carriers in Texas, with very few exceptions, no worker's compensation claims were included in the study. Patients with the following severe medical conditions anywhere in their medical history were excluded from the study: cancer, crushing injury, fractures, IV drug abuse, neurologic impairment, other acquired deformities, possible tumor/infection, pregnancy, rheumatoid arthritis, and trauma.

Ethics

This study used deidentified data and received institutional review board exemption status from the Baylor University Institutional Review Board.

Main Independent Variable

We examined costs and utilization among the following provider types: AP, chiropractor, orthopedic specialist, physical therapist, primary care provider, and physiatrist.

Control Variables

Our study controlled for age and gender and several variables to control for differences in patient acuity levels. We chose the Elixhauser Comorbidity Index (ECI)[16,17] as our primary measure of patient acuity. We also controlled for various patient characteristics based on diagnosis and procedure, all of which were binary: whether the primary diagnosis was a lower back pain (LBP) diagnosis, case complexity, smoking status, obesity, chronic pain, substance abuse disorder, mental health disorders, and prior spine surgery. Several control variables were modeled after a study examining entry setting and future lower back pain costs and utilization by Fritz et al.[18] For each episode, all claims 1 year prior to the index date were used to calculate the Elixhauser Comorbidity Index score and examined for the presence of each of our other patient acuity variables.[19,20]

Dependent Variables

We examined the association of provider type on total (allowable) costs per episode, utilization of advanced diagnostic imaging (eg, magnetic resonance imaging and computed tomography scans), utilization of neck- or back-related surgery, hospitalization, length of episode, and number of visits per episode. For total allowable costs, pharmacy costs were excluded because of incomplete pharmacy claims. When calculating total costs per episode, we removed all $0 cost episodes.

Data Source

Data were obtained through a data analytics company affiliated with the insurer.

Bias

Because patients were not randomly assigned to provider types, we used propensity scores using inverse probability weighting to control for selection bias.[21] Propensity scores were estimated using tree-based generalized boosted models.[22] In this model, covariates at baseline (age, gender, LBP primary diagnosis, ECI, smoking status, obesity, chronic pain comorbidity, substance use disorder, mental health comorbidity, prior spine surgery, and diagnosis complexity) were included to develop propensity scores for the provider category. Balance checks were performed using effect size (es) and Kolmogorov-Smirnov (ks) metrics.[23]

Study Size

Our initial dataset consisted of nearly 900 million claim line items. Because of this very large size, no power statistics were calculated. Our analyses demonstrated that we were able to detect differences with a probability of <.0001.

Quantitative Variables

Each episode was assigned a provider type based on the provider type with the highest number of visits in the episode. Ties in the highest number of visits were broken by the highest allowable cost.

Statistical Methods

After determining the propensity scores, average treatment effects were computed using inverse probability weighting of each patient using generalized linear regression with a log-link, gamma errors for continuous outcomes (cost, number of visits, and length of episode) because of the positive skew in the data and logistic regressions for binary outcomes (utilization of advanced diagnostic imaging, surgery, hospitalization). All models were fit in R with the twang package[24] used for propensity score general boosted models, the cobalt package to perform balance checks,[25] and the survey package to perform weighted regression analyses.[26]

Other Analyses

We conducted secondary analyses that excluded episodes that were less than $100 or 1 visit to examine costs and utilization without patients who did not seek care beyond their initial visit. All analyses were done in R.[27]

Results

Participants

Based on our inclusion/exclusion criteria, the sample comprised 645 799 patient episodes.

Descriptive Data

A summary of the study sample, overall and by provider type, is shown in Table 1. While there were similarities by provider type, clear differences ($P < .001$ using Kruskal-Wallis tests for continuous variables and χ^2 tests for categorical) existed that confirm the importance of propensity score weighting prior to comparing provider outcomes. In particular, the patients visiting AP tended to be slightly younger (45.94 vs 48.32 overall), less likely to smoke (0.6% vs 1.2%), and less obese (4.9% vs 5.6%). Half as many patients visiting AP had chronic pain comorbidities (0.3%), more had Elixhauser Comorbidity indices below 1 (94% vs 87%), and fewer had complex diagnostic codes (17% vs 32%). In contrast, more patients visiting AP had mental health comorbidities (2.4% vs 1.8%). Differences existed among other providers as well, in some cases "favoring" AP, but not always. For example, patients using chiropractors were less obese (4.7%), had fewer pain comorbidities (0.2%), and included a higher percentage with low Elixhauser scores (95%). Only substance use disorder was not statistically different across providers ($P = .11$).

Outcome Data

The key outcomes based on the sample data without propensity score weighting are presented in Table 2. Figure 1 shows the covariate balance, unweighted as well as weighted using both es and ks as stopping rules. In the unweighted data, the largest significant imbalance between groups was observed for age, Elixhauser Comorbidity score, complexity of the diagnosis, and whether the diagnosis was lower back or not. For the 2 least balanced variables, age and Elixhauser scores, AP had a younger sample

Table 1. *Characteristics of the Study Sample*

Characteristic	Overall N = 645 799	Airrosti Provider N = 13 246	Chiropractor N = 221 511	Orthopedic Specialist N = 31 356	Physiatrist N = 17 824	Primary Care Provider N = 321 872	Physical Therapist N = 39 990	*P* Value
Age, y (mean, SD)	48.32 (11.27)	45.94 (10.23)	46.29 (10.98)	51.32 (11.07)	51.36 (10.89)	49.17 (11.26)	49.87 (11.68)	<.001
Female	50%	48%	46%	47%	51%	52%	55%	<.001
Smoker	1.2%	0.6%	0.9%	1.5%	1.5%	1.4%	1.1%	<.001
LBP primary diagnosis	16%	12%	15%	31%	38%	14%	19%	<.001
Obesity	5.6%	4.9%	4.7%	6.1%	5.4%	6.2%	6.1%	<.001
Chronic pain comorbidity	0.6%	0.3%	0.2%	0.9%	4.2%	0.6%	0.7%	<.001
Substance use disorder	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	.11
Mental health comorbidity	1.8%	2.4%	1.6%	1.9%	1.7%	1.8%	1.9%	<.001
Prior spine surgery	<0.1%	0%	0%	0.8%	0%	0%	0%	<.001
Low Elixhauser (<1)	87%	94%	95%	90%	82%	81%	84%	<.001
Complex diagnostic code	32%	17%	23%	48%	52%	37%	24%	<.001

LBP, lower back pain.

with lower comorbidity before applying propensity score weighting. After weighting, there was a negligible difference between the groups.

Main Results

Table 3 shows the results with inverse propensity score weighting. The average episode cost for AP increased from $958 (unweighted) to $1471 (weighted). Costs for orthopedic specialists ($3478), physiatrists ($2346), and physical therapists ($4324) were all higher than those for AP ($P <$ .0001). AP cost was significantly higher than primary care provider cost ($503, $P <$ .0001). The AP cost estimate was also higher than that for chiropractors ($1336), but not by a statistically significant amount ($P =$ .343).

All provider types had larger SDs for average episode cost compared to AP (SD: $7415), except for primary care provider (SD: $5500), with the largest being physical therapist (SD: $20 464).

Chiropractors, physiatrists, and physical therapists had significantly more visits and longer episode lengths than AP ($P <$ .001); both metrics were statistically higher ($P <$ .001) than those for primary care providers. Orthopedic specialists were statistically higher in terms of length of episode ($P =$ .04) but lower in terms of number of visits ($P <$ .001) compared with AP, but in both cases, the estimates were more similar to AP (number of visits, 0.9 less; length of episode, 1.6 days longer).

The SDs of both average number of visits per episode and average episode length were larger for all provider types than for AP (visits: 3.02; episode length: 47.4). Among the 5 other provider types, primary care providers had the lowest SDs at 5.29 visits/49 days, and chiropractors had the highest SDs at 13.85 visits/155.4 days.

Orthopedic specialists and physical therapists had the highest odds of hospitalization compared with AP, with an estimated 5× odds ($P <$ .001; 95% CI: [3, 8.3] for orthopedic specialists and similar for physical therapists). The increase in odds compared with AP was not statistically significant for chiropractors ($P =$ .458) or primary care providers ($P =$ .866). Orthopedic specialists, physiatrists, and physical therapists also had increased odds of advanced diagnostic imaging or surgery during an episode compared with AP. For advanced diagnostic imaging, all 3 provider types had estimates of more than 4× odds compared with AP. For surgery, the odds were 2× as much for physiatrists and 5.5× and 6× as much for physical therapists and orthopedic specialists, respectively, than for AP. Primary care providers were significantly lower in terms of advanced diagnostic imaging ($P <$ .001; Odds Ratio (OR) = 0.79; 95% CI: [0.7, 0.9]) and surgery ($P =$ .0287; OR = 0.63; 95% CI: [0.4, 0.95]) compared with AP. Chiropractors

Table 2. *Unweighted Outcomes by Provider Type*

Outcome	Overall N = 645 799	Airrosti Provider N = 13 246	Chiropractor N = 221 511	Orthopedic Specialist N = 31 356	Physiatrist N = 17 824	Primary Care Provider N = 321 872	Physical Therapist N = 39 990
Cost	$1176.87	$957.81	$853.42	$4446.58	$3401.26	$495.26	$4972.03
Mean (SD)	($9726.19)	($3664.11)	($4888.01)	($24 259.79)	($16 308.98)	($5231.28)	($23 691.92)
Number of visits	4.68	3.54	7.97	3.00	5.95	1.68	11.73
Mean (SD)	(8.64)	(2.71)	(11.81)	(4.63)	(8.17)	(2.93)	(11.33)
Episode length	42.96	26.34	76.36	33.94	81.39	13.92	87.20
Mean (SD)	(103.57)	(42.27)	(144.06)	(71.22)	(139.29)	(48.73)	(101.08)
Hospitalization	0.9%	0.2%	0.3%	3.5%	1.9%	0.7%	3.3%
Imaging	8.7%	5.3%	4.5%	29.0%	31.0%	6.4%	27.0%
Surgery	1.0%	0.3%	0.5%	5.2%	2.2%	0.5%	4.3%

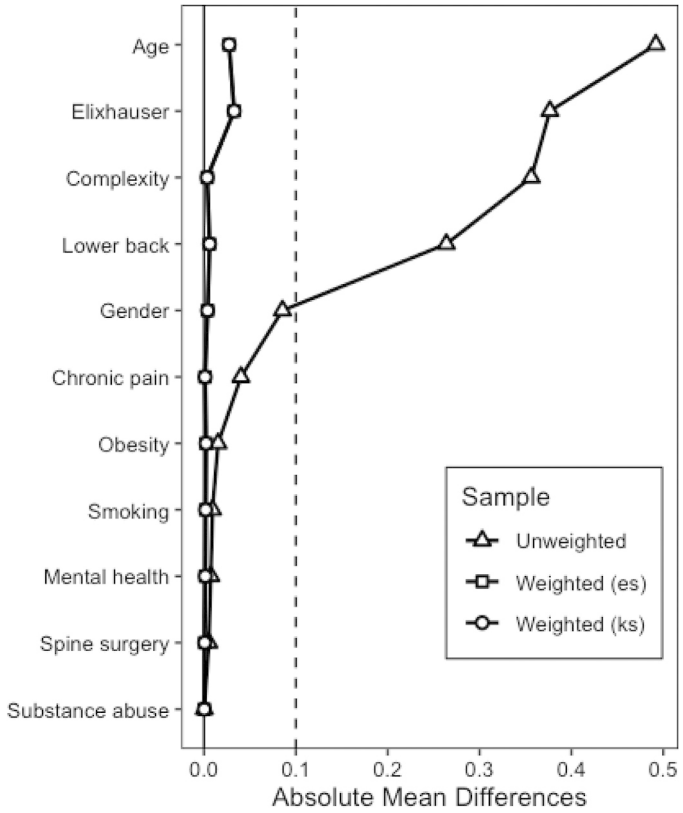

Fig 1. *Covariate balance maximum scores across provider pairs, unadjusted and propensity weighted (es and ks stopping rules). es,* effect size*; ks,* Kolmogorov-Smirnov.

Table 3. *Outcomes by Provider Type With Propensity Score Weights*

Outcome	Airrosti Provider N = 13 246	Chiropractor N = 221 511	Orthopedic Specialist N = 31 356	Physiatrist N = 17 824	Primary Care Provider N = 321 872	Physical Therapist N = 39 990
Cost	$1470.50	$1336.60	$3477.90	$2346.30	$503.00	$4324.20
Mean (SD)	($7414.97)	($7963.80)	($19 599.36)	($11 592.23)	($5500.00)	($20 464.49)
P value	—	.364	<.001	<.001	<.001	<.001
Number of visits	3.87	8.84	2.96	5.73	1.70	11.48
Mean (SD)	(3.1)	(12.9)	(4.6)	(8.1)	(3.0)	(11.0)
P value	—	<.001	<.001	<.001	<.001	<.001
Episode length	31.1	85.6	32.7	69.9	13.9	84.5
Mean (SD)	(47.4)	(155.4)	(71.9)	(122.0)	(49.0)	(97.5)
P value	—	<.001	<.05	<.001	<.001	<.001
Hospitalization	—	1.2	5.0	2.2	1.04	5.1
OR (95% CI)		(0.7, 2.0)	(3.0, 8.3)	(1.3, 3.7)	(0.6, 1.7)	(3.1, 8.4)
P value		.4580	<.001	<.01	.8659	<.001
Imaging	—	0.85	4.38	4.38	0.79	4.29
OR (95% CI)		(0.77, 0.94)	(4.0, 4.8)	(4.0, 4.8)	(0.7, 0.9)	(3.9, 4.7)
P value		<.001	<.001	<.001	<.001	<.001
Surgery	—	1.33	6.02	2.06	0.63	5.57
OR (95% CI)		(0.9, 2.0)	(4.0, 9.1)	(1.3, 3.2)	(0.4, 0.95)	(3.7, 8.4)
P value		.1796	<.001	<.001	<.05	<.001

OR, odds ratio.

were significantly lower in terms of advanced diagnostic imaging ($P < .001$; OR = 0.85, 95% CI: [0.77, 0.94]) but slightly higher in terms of surgery (OR = 1.33) than AP, a result that was not statistically significant ($P = .1796$).

Discussion

Key Results

We found cost differences between AP and NAP, except for chiropractors, whose episode costs were not statistically different. The largest cost difference was between AP and physical therapists (+$2854). AP costs were higher than primary care provider costs (+$968). In addition, except for primary care providers, the SDs for all NAP were greater than those for AP. Regarding our utilization measures, orthopedic specialists, physiatrists, and physical therapists had significantly higher number of visits, longer episode lengths, higher number of hospitalizations, higher use of advanced imaging, and higher use of surgery than AP, except for fewer visits for orthopedic specialists. AP had higher use of advanced imaging than chiropractors. Chiropractors had higher number of visits and longer episode lengths compared with AP, and there was no difference in hospitalization or use of surgery compared with AP. AP had higher levels of utilization across all measures compared with primary care providers, except for hospitalization, in which there was no difference. Except for primary care providers, the SDs for both episode length and visits were greater for all provider types compared with AP.

Our findings align with the results of several published studies comparing cost and utilization differences among provider types in the treatment of back pain. In a randomized control trial, Kominski et al[28] concluded that chiropractic care had higher mean outpatient costs and a higher number of visits than medical care. Shekelle et al[29] concluded that orthopedic physicians have the highest average cost per episode of care, general practitioners the lowest

cost, and chiropractic care the second lowest. In 2 studies comparing only chiropractic care and physical therapy, both studies concluded that chiropractic care was less expensive than physical therapy.[30,31] In comparing chiropractors with primary care doctors, Grieves et al[32] found primary care doctors to be lower in cost than chiropractors.

There were other studies, however, that differed from ours in their findings. In a prospective study comparing primary care providers, chiropractors, and orthopedic surgeons, Carey et al[33] reported that chiropractic care had the highest outpatient charges and the highest number of visits. In a retrospective study using claims data, Fritz et al[34] compared primary care, physiatry, chiropractic, and physical therapy and found that entry into chiropractic care and physical therapy settings resulted in the lowest episode costs. They also found that these entry settings were less likely to involve imaging, surgeon visits, or injections relative to medical providers. In numerous retrospective studies, the cost of chiropractic care was lower than medical care or nonchiropractic care.[28-38]

Interpretation

There are several plausible explanations for our results. The first is Airrosti's standardization of clinical practice through initial and ongoing training, ongoing and systematic quality assurance and utilization management, and strict adherence to clinical practice guidelines. The treatment pathways align with recommended clinical practice guidelines that stress the importance of nonpharmacologic treatments as a first step for lower back pain, and "discourages the use of diagnostic imaging, prescription medications, steroid injections, and spinal surgery as early care options."[39(p3)] Additional recommended treatment guidelines that Airrosti adheres to include: provider education and advice facilitated through a standard 1-hour visit length, for patients to remain active and stay at work by focusing on resuming daily activities of living and exercise routines as soon as possible, and exercise for patients with chronic back pain through Airrosti's prescribed at-home exercises.[5] Similarities in cost, hospitalization, and use of surgery between AP and chiropractors could be explained by Airrosti's employment of mostly licensed chiropractors and chiropractic clinical practice that follows similar treatment pathways.

The second possible explanation for AP lowered costs and utilization over several NAP types is the application of quality improvement practices emphasizing appropriateness of care. In particular, Airrosti employs a business policy and practice of not charging patients for the initial visit if the patient's condition is not considered appropriate for care. In these cases, patients are referred to a more appropriate provider type. This pattern of initial assessment and controlled escalation of care aligns with recently recommended treatment service models emphasizing risk stratification.[39,40]

Last, lowered costs and utilization compared to orthopedic specialists, physiatrists, and physical therapy may result from Airrosti's focus on a limited set of conditions assessed to respond favorably to its treatment modality; that is, manual therapy targeting distortions in connective tissue. For example, Airrosti treats 462 of approximately 4000 ICD-10 diagnosis codes related to back and neck. This limited scope of practice could enhance clinical outcomes resulting in fewer referrals to more expensive treatment options such as injections or surgery. In the parlance of Clay Christensen's disruptive innovation framework for health care,[41] this may be considered a value-adding process business for select musculoskeletal conditions.

The reduced costs and utilization of primary care providers compared to AP could be attributable to care-seeking tendencies of those using primary care providers. A study by Fritz and associates[7] found a significant number of people seeking care for lower back pain in primary care provider settings did not receive care beyond the initial entry visit. Our secondary analysis excluded episodes that were less than $100 or 1 visit and found that primary care provider costs were only slightly lower than AP (−$220) and with 0.64 fewer visits. In this analysis, the episode length, hospitalization, and use of advanced imaging were all higher for primary care providers than AP care with no difference in the use of surgery.

Generalizability

Our study has several important implications. In addition to generally lower costs and utilization for AP compared to most provider types, all provider types except for primary care providers had larger variations in cost, visits, and episode length. So, patients seeking care from NAP for back or neck conditions had higher costs, on average, and more uncertainty of the costs. In contrast, the majority of care costs by AP in the claims dataset we analyzed were simple multiples (×1, ×1.5, ×2, ×3, etc) of a base cost, suggesting transparency in the pricing model based on visits. In terms of utilization, patients treated by chiropractors, physiatrists, and physical therapists in our study required a higher number of office visits and received care over a longer period; treatment patterns that typically reduce employee productivity through absenteeism.

Improving clinical treatment pathways that are cost-effective among different provider types will require future research that combines total costs of care with quality and clinical outcome measures. The use of randomized clinical trials could allow for direct comparisons in cost-effectiveness between provider types.

Limitations

The use of claims data limited the comprehensive identification of diagnoses for each patient. Though additional diagnoses could have been gathered if our dataset was merged with the patient's electronic medical record, these data were

unavailable. Also, because of data limitations, our study did not include prescription medication costs that are known contributors to increasing costs associated with back and neck pain.[5] Additionally, the use of Texas data limits the generalizability of our findings outside the state. However, a recent study using 2020 U.S. Census data found Texas to be the second most diverse state in the country.[42] Furthermore, diagnostic imaging only included magnetic resonance imaging and computed tomography scans that were grouped together. Future research may examine the likelihood of individual types of diagnostic imaging by provider. As noted earlier, the main threat to the internal validity of our study is selection bias. Though we controlled for several patient demographic and acuity variables and achieved balance for all our covariates among our provider groups, there could be other unobserved differences among the patients that were not included in the propensity score adjustment. Last, our study did not include any patient outcome or quality of care indicators; thus, we could draw no conclusions regarding cost-effectiveness among the provider types.

Conclusion

We found financially significant cost and utilization differences between AP and NAP. Using patient episodes of care as the unit of analysis and propensity scoring to address the effects of selection bias, we found that orthopedic specialists, physiatrists, and physical therapists had higher costs and higher utilization compared with AP, with the exception of lower visits for orthopedic specialists compared with AP. Primary care providers had lower costs and utilization than AP, except for hospitalizations, in which no difference was found. AP were most similar to chiropractors in terms of costs and utilization; however, AP had higher use of advanced diagnostic imaging compared with chiropractors. Chiropractors had more visits and longer episodes compared with AP. These results suggest the potential benefits of increased emphasis and application of standardized treatment pathways to reduce variation and costs for spinal pain. This emphasis may have a profound impact on other providers of these defined services.

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Funding Sources and Conflicts of Interest

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Contributorship Information

Concept development (provided idea for the research): F.S.K.
Design (planned the methods to generate the results): F.S.K., D.J.K., N.S.F., R.X.S.
Supervision (provided oversight, responsible for organization and implementation, writing of the manuscript): F.S.K., D.J.K., R.X.S.
Data collection/processing (responsible for experiments, patient management, organization, or reporting data): D.J.K., R.X.S.
Analysis/interpretation (responsible for statistical analysis, evaluation, and presentation of the results): F.S.K., D.J.K., N.S.F., R.X.S., M.G., S.L.
Literature search (performed the literature search): F.S.K., T.H., S.L.
Writing (responsible for writing a substantive part of the manuscript): F.S.K., R.X.S.
Critical review (revised manuscript for intellectual content, this does not relate to spelling and grammar checking): F.S.K., D.J.K., N.S.F., R.X.S.

Practical Applications

- This study compared treatment expenditures and utilization of licensed doctors of chiropractic or doctors of physical therapy working for AP compared with NAP comprising 5 different provider types.
- Standard deviations for average episode cost, length, and number of visits were greater for all provider types compared with AP, except for primary care.
- AP providers using standardized treatment pathways had reduced costs, utlization, and variation in costs and utilization compared to most provider types for patients with spinal pain.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jmpt.2024.09.013.

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